Reactivity coefficients

How does changing one system parameter change reactivity?

21 LESSON

Read: FNRP § 9.2–9.4; the handwritten derivations and examples for Lesson 21.

Objective: Quantify reactivity changes due to fuel temperature, moderator temperature, and other parameters, including coolant void fraction.

1 From integral changes to local sensitivity

Last time: $\rho = (k-1)/k$, its numerical units, and the integral reactivity changes from shutdown to full power. Now hold the other independent parameters fixed and define

$$\boxed{\alpha_x = \left.\frac{\partial \rho}{\partial x}\right|_{\text{all other parameters fixed}}}.$$

For $x = T_F$, this is the **fuel temperature coefficient**, α_F (FTC). For $x = T_M$, it is the **moderator temperature coefficient**, α_M (MTC). Temperature coefficients are commonly reported in pcm/K. For boiling water reactors, the coolant void fraction (or steam quality) is another possible x.

Starting directly from the definition of ρ,

$$\frac{\partial \rho}{\partial k} = \frac{1}{k^2}, \qquad \alpha_x = \frac{1}{k^2}\frac{\partial k}{\partial x}.$$

Typically, coefficients are reported for near-critical states, or

$$\alpha_x \simeq \frac{1}{k}\frac{\partial k}{\partial x} \qquad (k \simeq 1).$$

Away from $k = 1$, the two normalizations differ.

2 A Finite Difference Approximation

Suppose we modeled a PWR fuel pin at two different moderator temperatures using OpenMC and found the following multiplication factors:

$$\underbrace{k_{600} = 1.28397}_{\text{nominal}}, \qquad \underbrace{k_{610} = 1.27228}_{\text{perturbed}}.$$

To estimate α_M at the nominal $T_M = 600$ K, we can apply a **finite-difference** approximation, or

$$\alpha_M \approx \left.\frac{\partial k}{\partial T_M}\right|_{600} \simeq \frac{k_{610} - k_{600}}{610 - 600} = -0.001169\ \text{K}^{-1},$$

or, in the more common units,

$$\boxed{\alpha_M \simeq \frac{-0.001169}{1.28397} = -9.1046 \times 10^{-4}\ \text{K}^{-1} \simeq -91\ \text{pcm/K}.}$$

Assumption check

Using $\rho = 1 - 1/k$ at both temperatures instead gives

$$\frac{\rho_{610} - \rho_{600}}{10} = \frac{k_{610} - k_{600}}{10 k_{600} k_{610}} = -71.6\ \text{pcm/K}.$$

Because $k \simeq 1.28$, this differs materially from the near-critical result; the difference is not rounding! Unless directed otherwise, the near-critical approximation should be used for reporting reactivity coefficients.

3 Expressing α_x with the Four Factors

Recall $k_\infty = \eta_T f p \varepsilon$. The handwritten η is written η_T here, following the slide notation. The product rule gives

$$\frac{\partial k_\infty}{\partial x} = f p \varepsilon \frac{\partial \eta_T}{\partial x} + \eta_T p \varepsilon \frac{\partial f}{\partial x} + \eta_T f \varepsilon \frac{\partial p}{\partial x} + \eta_T f p \frac{\partial \varepsilon}{\partial x}.$$

Divide by $k_\infty = \eta_T f p \varepsilon$:

$$\frac{1}{k_\infty}\frac{\partial k_\infty}{\partial x} = \frac{1}{\eta_T}\frac{\partial \eta_T}{\partial x} + \frac{1}{f}\frac{\partial f}{\partial x} + \frac{1}{p}\frac{\partial p}{\partial x} + \frac{1}{\varepsilon}\frac{\partial \varepsilon}{\partial x}.$$

Thus, in the near-critical approximation used throughout the handwritten four-factor analysis,

$$\boxed{\alpha_x \simeq \frac{1}{\eta_T}\frac{\partial \eta_T}{\partial x} + \frac{1}{f}\frac{\partial f}{\partial x} + \frac{1}{p}\frac{\partial p}{\partial x} + \frac{1}{\varepsilon}\frac{\partial \varepsilon}{\partial x}.}$$

Each term is the fractional sensitivity of one factor. The exact derivative of ρ_∞ is $1/k_\infty$ times this sum.

4 Recall the UO_2 resonance model

For a UO_2 fuel element,

$$p = \exp\left[-\frac{V_f N_{fe} I}{V_m \xi^m \Sigma_s^m}\right].$$

Here V_f and V_m are the fuel and moderator volumes, N_{fe} is the fertile-nuclide number density in the fuel, and $\xi^m \Sigma_s^m$ is the moderator slowing-down power. The effective heterogeneous resonance integral is I.

With ρ_f the fuel mass density and D the fuel diameter, define

$$I(300)\,[\text{b}] = 4.45 + 26.6\sqrt{\frac{4}{\rho_f D}},$$

$$I(T_F) = I(300)\left[1 + \tilde{\gamma}\left(\sqrt{T_F} - \sqrt{300}\right)\right],$$

$$\tilde{\gamma}\,[\text{K}^{-1/2}] = 0.0061 + 0.0047\frac{4}{\rho_f D}.$$

Insert ρ_f in g/cm^3, D in cm, and temperatures in kelvin. The subscript on ρ_f distinguishes mass density from reactivity. These relations are equivalent to FNRP Eq. (4.40), Table 4.3, Eqs. (9.13)–(9.14), and Table 9.1.

At fixed moderator conditions, the explicit T_F dependence in this simplified model enters through $I(T_F)$ and hence p. Moderator temperature affects moderator density, so it changes both p and f. The next page differentiates those pathways separately.

Lesson 21 | Fuel and moderator temperature feedback

5 Derive the fuel temperature coefficient

With the moderator and geometry fixed, write

$$C_F = \frac{V_f N_{fe}}{V_m \xi^m \bar{\Sigma}_s^m}, \qquad \ln p = -C_F I(T_F).$$

Since p is the only factor with explicit T_F dependence in this model,

$$\alpha_F \simeq \frac{1}{p}\frac{\partial p}{\partial T_F} = -C_F \frac{\partial I}{\partial T_F}$$
$$= -\ln\left(\frac{1}{p}\right)\frac{1}{I}\frac{\partial I}{\partial T_F}.$$

The empirical relation gives

$$\frac{\partial I}{\partial T_F} = I(300)\frac{\tilde{\gamma}}{2\sqrt{T_F}}, \qquad C_F I(300) = \ln\left[\frac{1}{p(300)}\right].$$

Therefore

$$\boxed{\alpha_F \simeq -\ln\left[\frac{1}{p(300)}\right]\frac{\tilde{\gamma}}{2\sqrt{T_F}}.}$$

The negative sign follows from $0 < p(300) < 1$ and $\tilde{\gamma} > 0$. Typical values for α_T are $\alpha_F \simeq -(2\text{–}3)$ pcm/K for PWRs and about -10 pcm/K for our TRIGA reactor.

Why does I depend on T_F? **Doppler broadening**!

Example: Fuel Temperature Coefficient

For $p(300) = 0.8$, compute α_F at $T_F = 1200$ K for a UO_2 fuel element with a diameter of 1 cm and mass density of 10 g/cm^3.

$$\tilde{\gamma} = 0.0061 + 0.0047\frac{4}{(10)(1)} = 0.00798 \text{ K}^{-1/2},$$

$$\alpha_F(1200) \simeq -\ln(1/0.8)\frac{0.00798}{2\sqrt{1200}}$$
$$= -2.57\times10^{-5} \text{ K}^{-1} = -2.57 \text{ pcm/K} \approx \boxed{-2.6 \text{ pcm/K}}.$$

6 Derive the moderator temperature coefficient

Recall

$$f = \frac{1}{1 + \varsigma\dfrac{V_m}{V_f}\dfrac{\bar{\Sigma}_{a,2}^M}{\bar{\Sigma}_{a,2}^F}}.$$

In the density-only model, T_M changes the moderator number densities: decreasing $\rho_m = \rho_{H_2O}$ lowers both $\bar{\Sigma}_{a,2}^M$ and $\xi^m\bar{\Sigma}_s^m$.
Define the volumetric thermal-expansion coefficient by

$$\beta_M = -\frac{1}{\rho_m}\frac{d\rho_m}{dT_M}.$$

Careful: this β_M is distinct from the delayed neutron fraction β.

Thermal utilization. Collect the fixed quantities into C and write $f = [1 + C\rho_m(T_M)]^{-1}$. Differentiating gives

$$\frac{1}{f}\frac{df}{dT_M} = -\frac{C\rho_m}{1+C\rho_m}\frac{1}{\rho_m}\frac{d\rho_m}{dT_M}$$
$$= (1-f)\beta_M > 0.$$

Resonance escape. Write $p = \exp[-A/\rho_m(T_M)]$, with A fixed. Then

$$\frac{1}{p}\frac{dp}{dT_M} = \frac{A}{\rho_m}\frac{1}{\rho_m}\frac{d\rho_m}{dT_M}$$
$$= -\beta_M \ln(1/p) < 0.$$

Combine these two contributions:

$$\boxed{\alpha_M \simeq -\beta_M\big[\ln(1/p) - (1-f)\big].}$$

The **overall sign depends on their relative magnitudes**. In this model, negative moderator feedback requires $\ln(1/p) > (1-f)$.

Example: Moderator Temperature Coefficient

For $p = 0.6$, $f = 0.95$, and $\beta_M = 2.9\times10^{-4}$ K^{-1},

$$\alpha_M \simeq -2.9\times10^{-4}\big[\ln(1/0.6) - (1-0.95)\big]$$
$$= -1.3364\times10^{-4} \text{ K}^{-1}$$
$$= -13.36 \text{ pcm/K} \approx \boxed{-13 \text{ pcm/K}}.$$

The resonance-escape penalty outweighs the utilization gain for these values.